Kafka

Vedio link : https://www.youtube.com/watch?v=clouToNoxGM&list=PLVz2XdJiJQxwoGuQb8lR-sTV26dz1SoXo

https://github.com/Java-Techie-jt/kafka-publisher

The Saga design pattern is a way to manage data consistency across microservices in distributed transaction scenarios. A saga is a sequence of transactions that updates each service and publishes a message or event to trigger the next transaction step.

Q. What is Kafka?

Ans. Kafka is an opensource distributed event streaming platform.

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Q: Why use Replication in Kafka?

Replication in Kafka is primarily used to ensure high availability, data durability, and fault tolerance by creating multiple copies of topic partition data across different brokers. It guarantees that if one broker fails, data remains accessible from other replicas, preventing data loss and allowing for seamless, automatic failover to a new leader.

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Q: can we run producer consumer without zookeeper in kafka?

Yes, you can run Kafka producers and consumers without ZooKeeper using Kafka Raft Metadata mode (KRaft). KRaft is an experimental feature that's not recommended for production use.

How it works

- **In KRaft mode, brokers use a consensus protocol called Raft to communicate with each other.**
- **Raft allows brokers to elect a leader and replicate metadata across the cluster.**
- **The quorum controller stores the state of the cluster using an event-sourced storage model.**
- **This ensures that the internal state machines can be recreated accurately.**

Benefits

- **KRaft mode simplifies Kafka management and deployment.**
- **It enhances scalability and fault tolerance.**
- **It can accommodate more partitions per cluster.**

Considerations

- **KRaft mode is experimental and should not be used in production.**

- **Some features, like ACLs, security features, transactions, partition reassignment, and JBOD, are not available in KRaft mode.**
- **You need to carefully monitor the broker's health and performance.**

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Q. What is ZooKeeper?

Ans. ZooKeeper: ZooKeeper is a centralized service for maintaining configuration information, naming, providing distributed synchronization, and providing group services. All of these kinds of services are used in some form or another by distributed applications. Each time they are implemented there is a lot of work that goes into fixing the bugs and race conditions that are inevitable. Because of the difficulty of implementing these kinds of services, applications initially usually skimp on them, which make them brittle in the presence of change and difficult to manage. Even when done correctly, different implementations of these services lead to management complexity when the applications are deployed.

Kafka Broker: Just a server. It is just an intermediate entity that helps in message exchanges between consumers and producers.

Kafka cluster: There can be one or more brokers in kafka cluster.

Topic: It specifies the category of messages or classification of messages. Listeners can then just respond to the messages that belong to the topics they are listening to.

Kafka download Link

https://kafka.apache.org/downloads

https://www.tutorialspoint.com/apache_kafka/apache_kafka_basic_operations.htm

Start zookeeper.start

D:\Software\java setup\kafka_2.12-1.1.0\bin\windows>zookeeper-server-start.bat D:\Software\java setup\kafka_2.12-1.1.0\config\zookeeper.properties

Start Kafka server

D:\Software\java setup\kafka_2.12-1.1.0\bin\windows>kafka-server-start.bat

D:\Software\java setup\kafka_2.12-1.1.0\config>server.properties

Create topics:

D:\Software\java setup\kafka_2.12-1.1.0\bin\windows>kafka-topics.bat --create --zookeeper localhost:2181 --replication-factor 1 –partitions 1 -topic <topic_name>

Create topic syntax:

--create --zookeeper localhost:2181 --replication-factor 1 –partitions 1 -topic <topic_name>

Test:

Produce Message in kafka:

D:\Software\java setup\kafka_2.12-1.1.0\bin\windows> kafka-console-producer.bat --broker-list localhost:9092 -topic <topic_name>

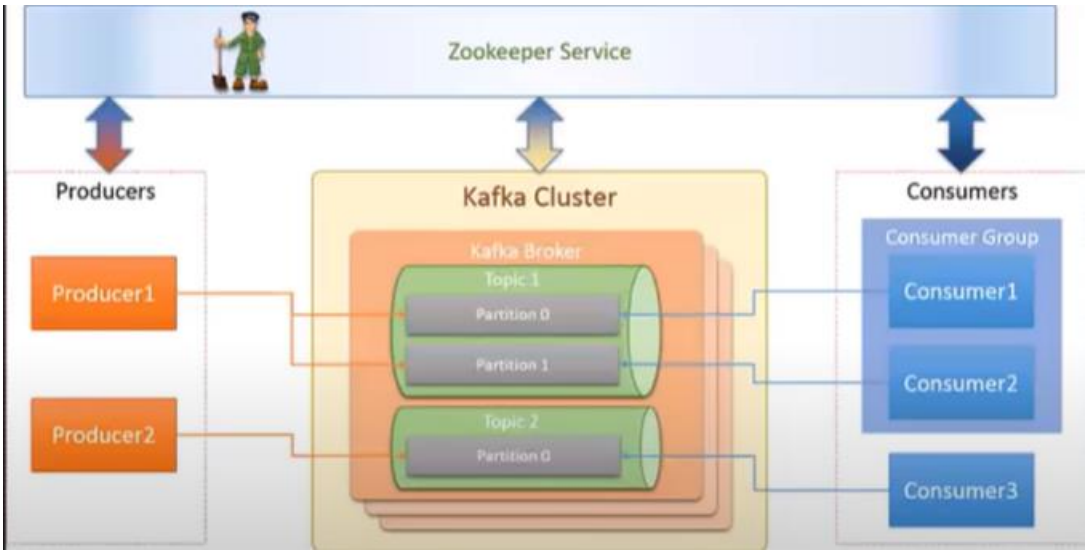
Consume a message in Kafka:

D:\Software\java setup\kafka_2.12-1.1.0\bin\windows> kafka-console-consumer.bat --zookeeper localhost:2181 -topic <topic_name>

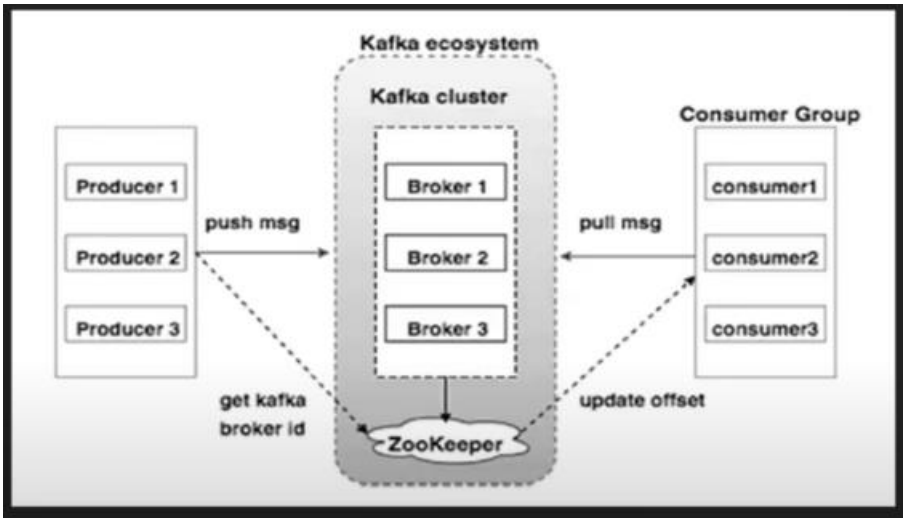
New for consumer:

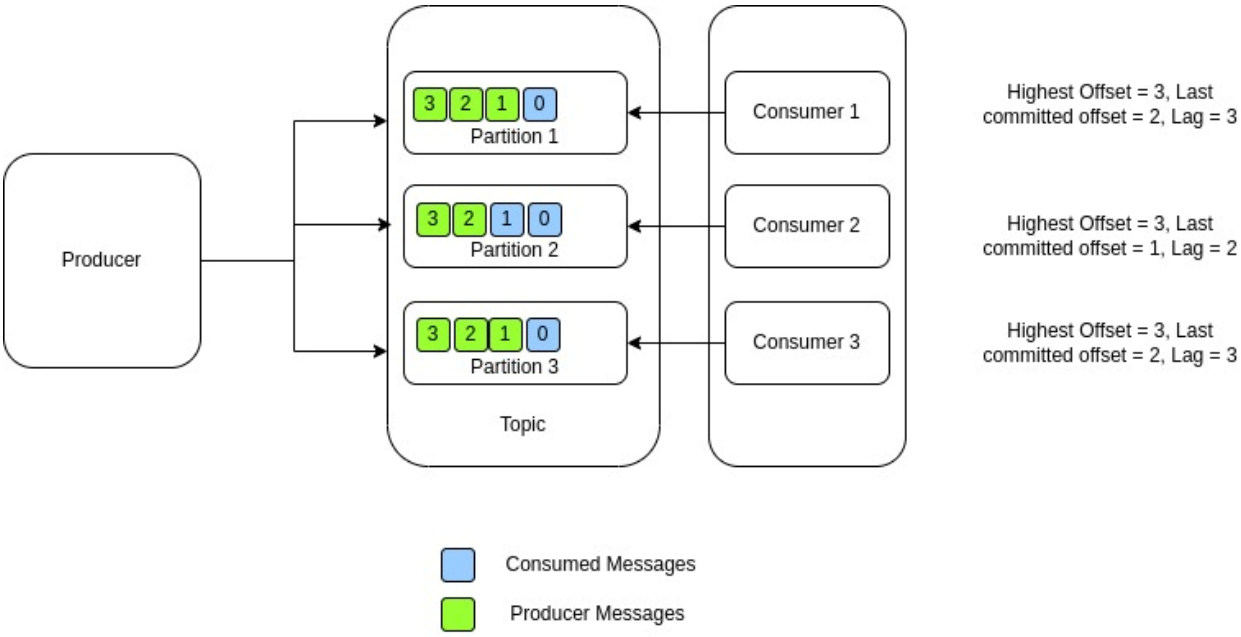
D:\Software\java setup\kafka_2.12-1.1.0\bin\windows>kafka-console-consumer.bat -bootstrap-server localhost:9092 --topic <topic_name>

Note: Bootstrap server is the Kafka server, having port number=9092.



Replication: It is nothing shadow instance of your streaming data.





What is Apache Kafka? https://www.javatpoint.com/apache-kafka

Apache Kafka is an open-source stream-processing software platform that is used to handle real-time data storage. It works as a broker between two parties, i.e., a sender and a receiver. It can handle trillions of data events in a day.

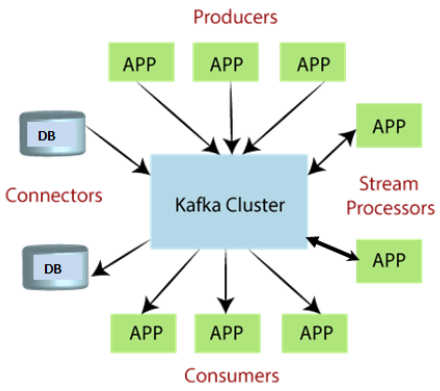
Apache Kafka is a software platform that is based on a distributed streaming process. It is a publish-subscribe messaging system that lets exchanging of data between applications, servers, and processors as well. Apache Kafka was originally developed by **LinkedIn**, and later it was donated to the Apache

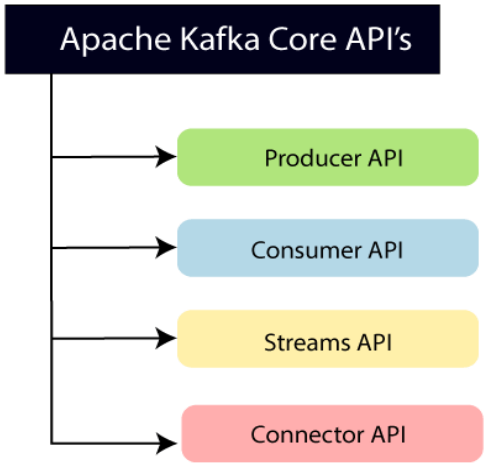
Software Foundation. Currently, it is maintained by **Confluent** under Apache Software Foundation. Apache Kafka has resolved the lethargic trouble of data communication between a sender and a receiver.

What is the Streaming process?

A streaming process is the processing of data in parallelly connected systems. This process allows different applications to limit the parallel execution of the data, where one record executes without waiting for the output of the previous record. Therefore, a distributed streaming platform enables the user to simplify the task of the streaming process and parallel execution. Therefore, a streaming platform in Kafka has the following key capabilities:

- As soon as the streams of records occur, it processes it.
- It works similar to an enterprise messaging system where it publishes and subscribes streams of records.
- It stores the streams of records in a fault-tolerant durable way.





Producer API: This API allows/permits an application to publish streams of records to one or more topics.

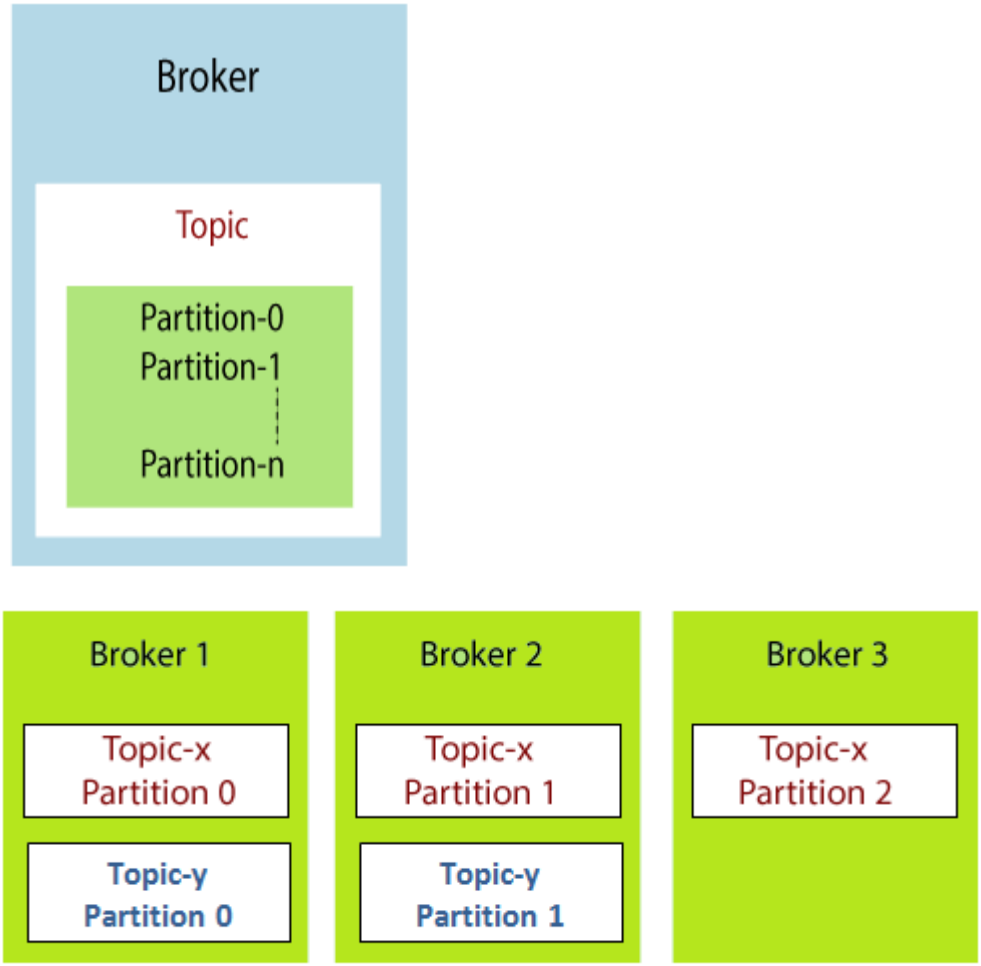
Consumer API: This API allows an application to subscribe to one or more topics and process the stream of records produced to them.

Streams API: This API allows an application to effectively transform the input streams to output streams. It permits an application to act as a stream processor which consumes an input stream from one or more topics and produces an output stream to one or more output topics.

Connector API: This API executes the reusable producer and consumer APIs with the existing data systems or applications.

Why Apache Kafka

1. Apache Kafka is capable of handling millions of data or messages per second.
2. Apache Kafka works as a mediator between the source system and the target system. Thus, the source system (producer) data is sent to Apache Kafka, where it decouples the data, and the target system (consumer) consumes the data from Kafka.
3. Apache Kafka is having extremely high performance, i.e., it has a really low latency value of less than 10ms which proves it as a well-versed software.
4. Apache Kafka has a resilient architecture that has resolved unusual complications in data sharing.
5. Organizations such as NETFLIX, UBER, Walmart, etc., and over thousands of such firms make use of Apache Kafka.
6. Apache Kafka is able to maintain the fault-tolerance. Fault tolerance means that sometimes a consumer successfully consumes the message that was delivered by the producer. But, the consumer fails to process the message back due to backend database failure, or due to the presence of a bug in the consumer code. In such a situation, the consumer is unable to consume the message again. Consequently, Apache Kafka has resolved the problem by reprocessing the data.
7. Learning Kafka is a good source of income. So, those who wish to raise their income in the future in the IT sector can learn.



Each broker is holding a topic, namely Topic-x with three partitions 0,1 and 2. Remember, all partitions do not belong to one broker only, it is always distributed among each broker (depending on the quantity). Broker 1 and Broker 2 contain another 'topic-y' having two partitions 0 and 1. Thus, Broker 3 does not hold any data from 'topic-y'. It is also concluded that no relationship ever exists between the broker number and the partition number.

Kafka Topic Replication

When to Use RabbitMQ vs Kafka

To summarize, if you're looking for a message broker to handle high throughput and provide access to stream history, Kafka is the likely the better choice. If you have complex routing needs and want a built-in GUI to monitor the broker, then RabbitMQ might be best for your application.

Redis is another message broker option. Being in-memory only, Redis is faster than even Kafka. It works best for customers whose destination can receive data far faster than the data can be generated.

Differences between RabbitMQ and Apache Kakka?

RabbitMQ employs the smart broker/dumb consumer model. The broker consistently delivers messages to consumers and keeps track of their status. Kafka uses the dumb broker/smart consumer model. Kafka doesn't monitor the messages each user has read.

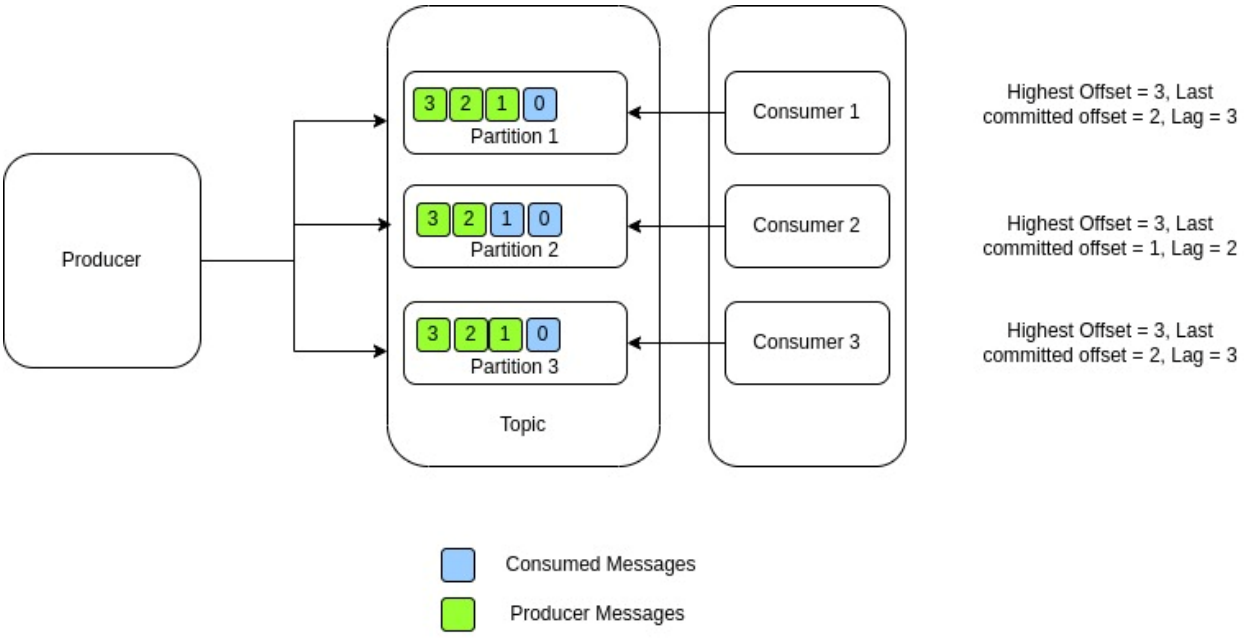
Differences between RabbitMQ and Apache Kakka in a tabular form:

Parameter	RabbitMQ	Kafka
Performance	Up to 10K messages per second	Up to 1 million messages per second
Data Type	Transactional	Operational
Synchronicity of messages	Can be synchronous/asynchronous	Durable message store that can replay messages
Topology	Exchange type: Direct, Fan out, Topic, Header-based	Publish/subscribe based
Payload Size	No constraints	Default 1MB limit
Usage Cases	Simple use cases	Massive data/high throughput cases
Data Flow	Distinct bounded data packets in the form of messages	Unbounded continuous data in the form of key-value pairs.
Data Unit	Message	Continuous stream
Data Tracking	Broker/Publisher keeps track of message status (read/unread)	Broker/Publisher keeps only unread messages; it doesn't retain sent messages.
Broker/Publisher Type	Smart	Dumb
Consumer Type	Dumb	Smart
Routing messages	Complex routing is possible based on event types	Complex routing is not possible; however, we can subscribe to individual topics.
Topology	Exchange queue topology	publish/subscribe topology
Message delivery system	Message pushed to specific queues	Pull-based model; consumer pulls messages as required
Message management	Prioritize messages	Order/Retain/Guarantee messages

Parameter	RabbitMQ	Kafka
Message Retention	Acknowledgment based	Policy-based (e.g., ten days)
Event storage structure	Queue	Logs
Consumer Queues	Decoupled Consumer queues	Coupled consumer partition/groups

Q: Kafka consumer lag? https://www.redpanda.com/guides/kafka-performance-kafka-consumer-lag
Ans: Kafka consumer lag is the difference between the last message produced by the producer and the offset committed by the consumer group. It represents the consumer processing delay.

Kafka consumer lag concept	Description
What is Kafka consumer lag?	Kafka consumer lag is the difference between the last offset stored by the broker and the last committed offset for that partition.
Reasons for Kafka consumer lag	Four common reasons for consumer lag are (1) Incoming traffic surges, (2) Data skew in partitions, (3) Slow processing jobs, and (4) Errors in code and pipeline components
Monitoring Kafka consumer lag	Teams can monitor Kafka consumer lag with the consumer group script, Burrow (a Kafka monitoring companion), or generic monitoring tools and exporters.
Strategies for addressing Kafka consumer lag	The best strategy to address Kafka consumer lag will vary depending on the underlying cause. Processing logic optimizations, partition count modifications, rate limiting, and configuration tuning are four common strategies to address consumer lag.



Key Concepts in Kafka

Apache Kafka is a *publish-subscribe* messaging system which lets you send messages between processes, applications, and servers. Applications which need to read data from Kafka use a KafkaConsumer to subscribe to Kafka topics and receive messages from these topics.

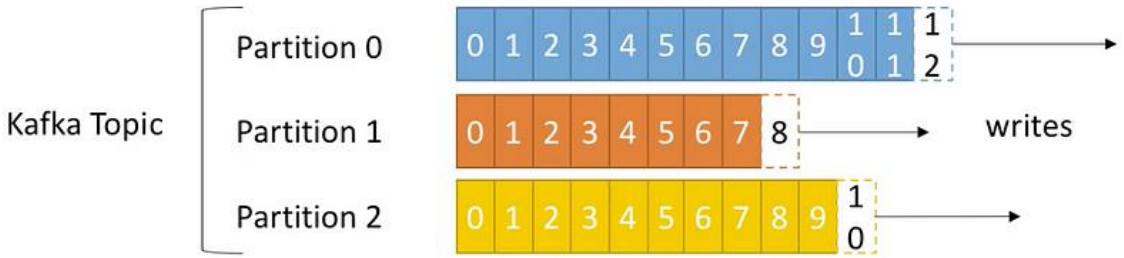
Reading data from Kafka is a bit different than reading data from other messaging systems, and there are few unique concepts involved. *Let's first review some of the key concepts in Kafka.*

1.Topics

Similar to a table in database, topics is a particular stream of data. Topics consist of one or more partitions, ordered, immutable sequences of messages to which Kafka appends new messages.

2. Partitions and Offset

Topics are split in partitions. Each message within a partition gets an incremental id, called offset.

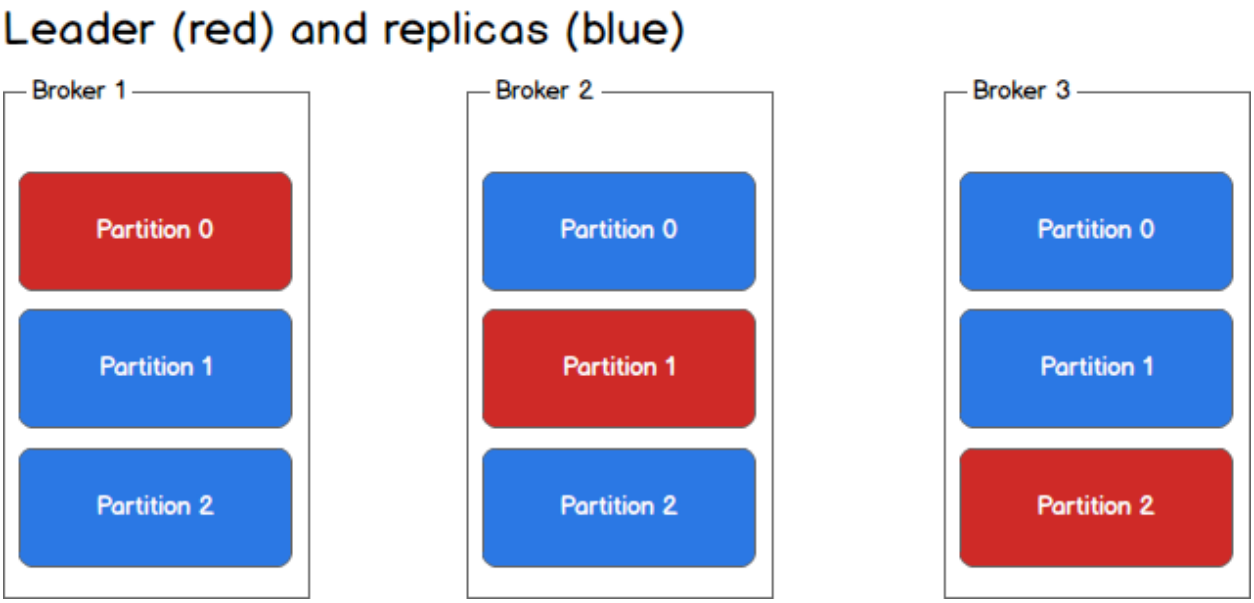


3. Brokers

A Kafka cluster consists of one or more brokers. Partitions are spread across these brokers. After connecting to any broker, you will be connected to the entire cluster.

4.Replicas

Topics have a replication factor to make sure if one broker is down, another broker can serve the data.



5. Zookeeper

Zookeeper manages brokers, helps in performing leader election for the partition.

6. Producer

Producers write data to topics.

7. Consumer

Consumer read data from a topic.

8. Lag

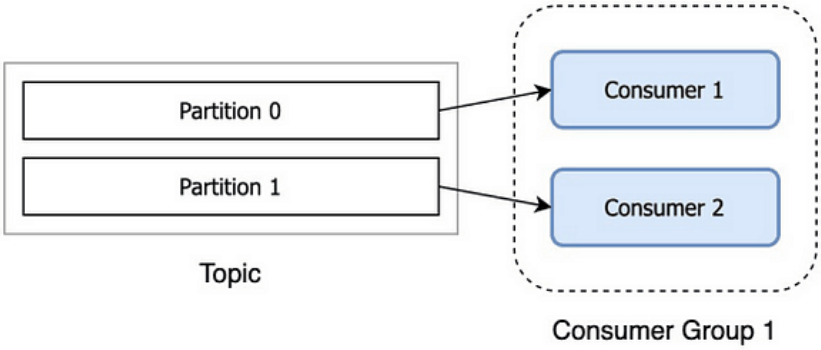
A consumer is lagging when it's unable to read from a partition as fast as messages are produced to it. Lag is expressed as the number of offsets that are behind the head of the partition.

The time required to recover from lag depends on how quickly the consumer is able to consume messages per second:

recover time in seconds = messages / (consume message per second - produce message per second)

What is Consumer Group?

Consumer can be grouped together for a given topic for **maximizing read throughput**. Each consumer in a group read from mutually exclusive partitions.

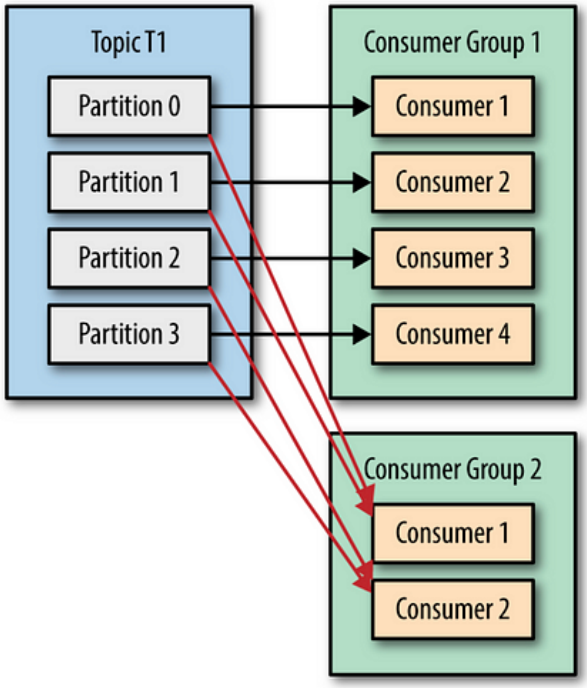


If all the consumer instances have the same consumer group id, then the records will effectively be load-balanced over the consumer instances and each consumer in the group will receive messages from a different subset of the partitions in the topic.

When to use different Consumer Groups?

When the consumers are performing different operations on the same topics, we should use different consumer groups.

For example, some consumers might update the downstream databases, while other consumers might do some real-time analysis with the data. In this case, we should register these consumers with different group-ids.

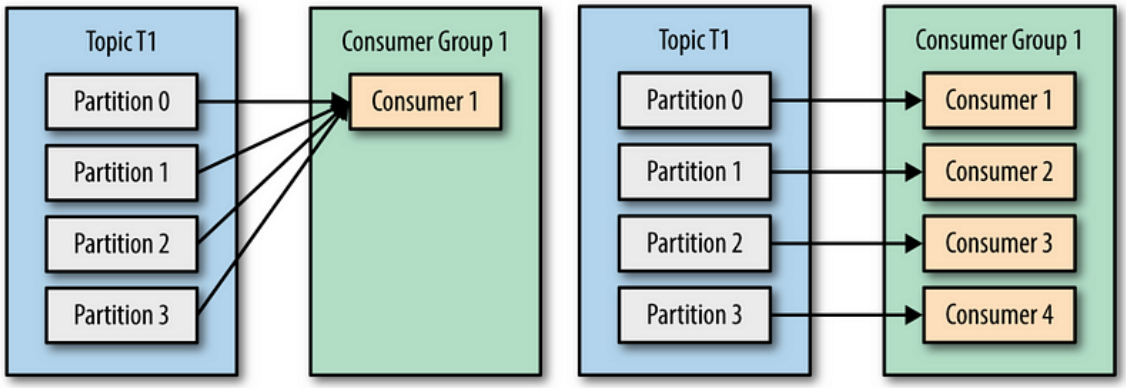


Why we need consumer parallel processing?

If you have a topic with four partitions and only one consumer in a group, that consumer would consume records from all the partitions.

This may work for you in some cases, but what if the rate at which producers write messages to the topic exceeds the rate at which your consumer application can process them? If you are still limited to a single consumer reading and processing the data, your application may be unable to keep up with the rate of incoming messages. In Kinesis, this latency is called iterator age. Obviously there is a need to scale consumer consumption from topics. Just like multiple producers can write to the same topic, we need a similar mechanism to allow multiple consumers to read from the same topic, sharing the data between them.

Scale data consumption horizontally

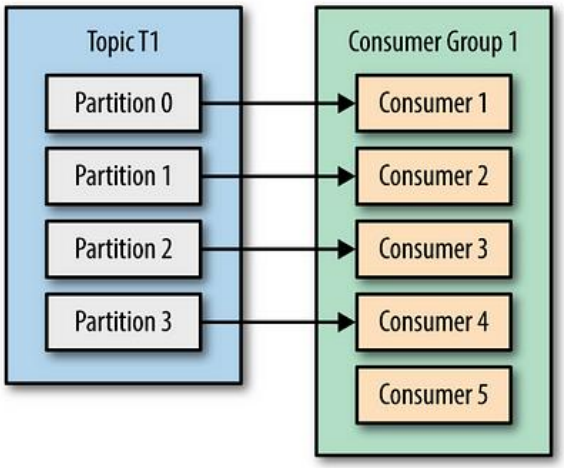


Scale horizontally

The main way we scale data consumption from a Kafka topic is by adding more consumers to the consumer group. It is a common operation for Kafka consumers to do high-latency operations such as writing to databases or a time-consuming computation.

If you know that you will need many consumers to parallelize the processing, you can plan accordingly with the number of partitions to parallel the consumer works.

It’s also important not to have more consumers in a group than you have partitions. The extra consumers will just sit idle, since all the partitions are taken.



Consumer 5 is idle

Partition Rebalance

what will happen if we add a new consumer to the group? The new consumer will start consuming messages from partitions previously consumed by another consumer. If it leaves the group, the partitions it used to consume will be consumed by one of the remaining consumers.

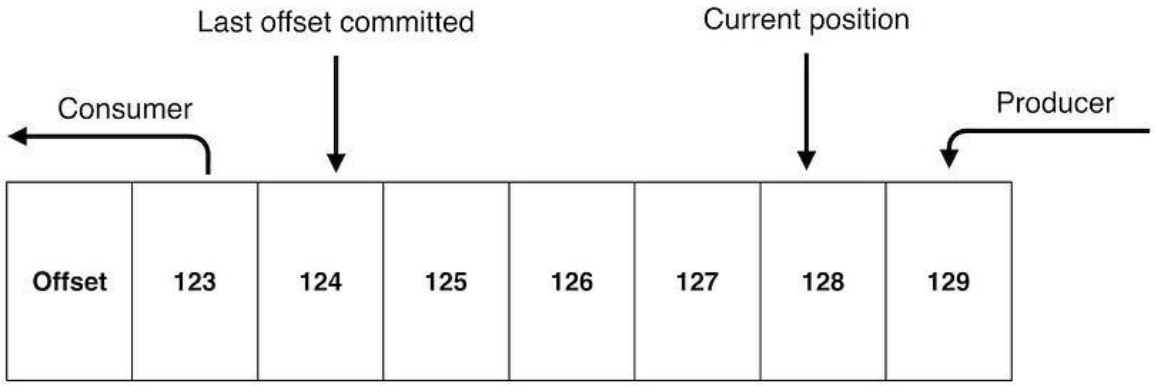
We call moving partition ownership from one consumer to another a *rebalance*.

Commits and Offsets

Current offset is the offset from which next new record will be fetched.

Committed offsets is the last committed offset for the given partition. Consumer produces a message to Kafka, to a special __consumer_offsets topic, with the committed offset for each partition.

If a consumer crashes or a new consumer joins the consumer group, this will *trigger a rebalance.* In order to know where to pick up the work, the consumer will read the latest committed offset of each partition and continue from there. We call this action of updating the current position in the Kafka partition a commit.



Vertical Scaling

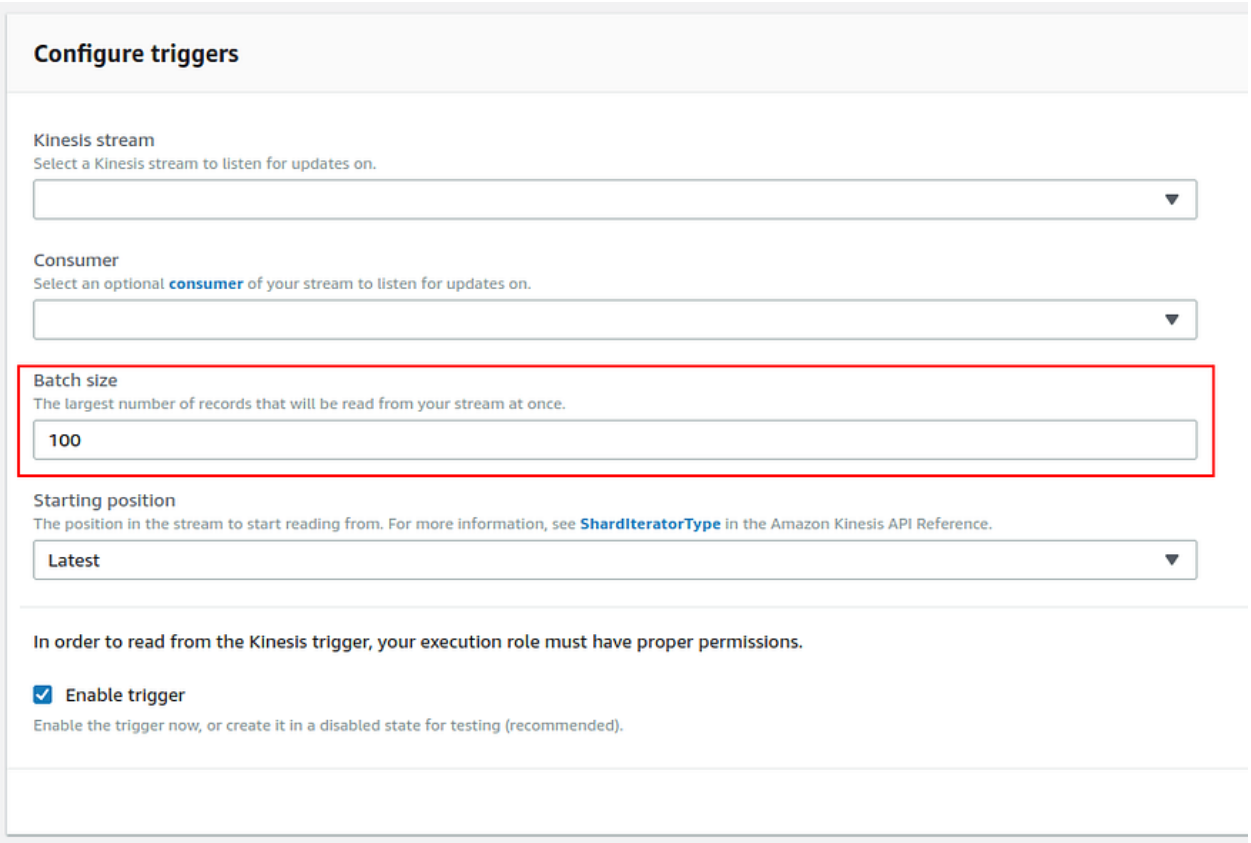
What if horizontally scaling is not enough or our consumer number already reaches the number of partitions? If you are dealing with high volume data or your consumer is doing expensive calculation, this can be a common question for you.

Single-threaded model vs multi-threaded model

A typical single-threaded model could be: first, it retrieves records using the poll() method, and then it processing fetched records.

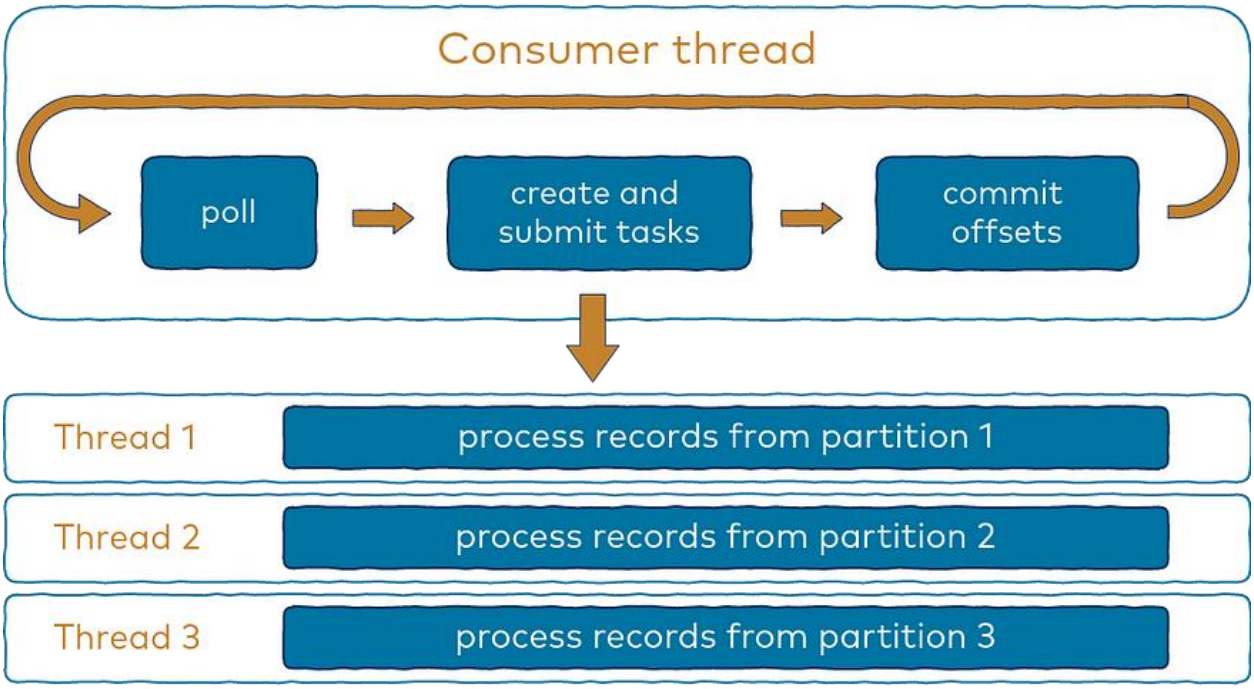
Because records are fetched and processed by the same thread, they are processed in the same order as they were written to the partition. This ensures the processing order guarantees.

For example, in AWS Kinesis with Lambda architecture, you can change the batch size which controls the maximum number of records that can be sent to your Lambda function with each invoke.



increase the Lambda batch size will increase throughput

***Multithreading* is "the ability of a central processing unit (CPU) (or a single core in a multi-core processor) to provide multiple threads of execution concurrently, supported by the operating system." In situations where the work can be divided into smaller units, which can be run in parallel, without negative effects on data consistency, multithreading can be used to improve application performance.**



a multi-threaded model

A multi-threaded model may be processing each message in a separate thread taken from a thread pool, while using automatic offset commits. However, a Multi-threaded model may cause some undesirable effects:

- **Offset might be committed before a record is processed by consumers**
- **Message processing order can't be guaranteed since messages from the same partition could be processed in parallel**

Multi-threaded model is useful when the other system can accept such high load and when there is no dependency on the order of processing.

Conclusion